

Bryan Zang

 bryanzang |  bryan-zang |  bzang19@outlook.com |  (647)-393-9656

SKILLS

Programming

- Python, SQL, R, MATLAB, NumPy, Pandas, PySpark, scikit-learn, TensorFlow, Julia, Java, C, LaTeX

Technologies

- Databricks, Excel, Fabric, PowerBI, Jupyter, Git, Refinitiv, Bloomberg Terminal, Power Automate

WORK EXPERIENCE

Research Assistant — Bank of Canada, Operational Risk and Resilience Analysis

Apr 2026 – Present

- Analyzed **market-risk** indicators using the Canadian Overnight Repo Rate Average in Databricks and PowerBI to support inter-active **risk monitoring** and model analysis.
- Streamlined **big data ingestion frameworks** from multiple internal warehouses in Fabric to ensure robustness of data availability.
- Developed **agentic analytical tools** through Copilot studio to assess **model risk** documentation and **compliance controls** for the Bank's Market Data Management research curve.
- Documented and validated structural tests on **cross-currency swap operations** collaborating with 5 central banks to maintain the consistency and transparency of market workflows.

Research Assistant — Bank of Canada, Debt Management

Jul 2025 – Present

- Analyzed **large financial-market datasets** using Pyspark, Databricks, and PowerBI to support research on [Government of Canada debt](#), [market well-being](#), investor behavior, policy reporting, and [debt-distribution framework revisions](#).
- Developed a daily volume-weighted bond-specialness indicator using Python and security-level transaction data to measure **relative demand and scarcity** in the Government of Canada bond market.
- Reduced runtime efficiency of data pipelines by **100 fold** using code parallelization and data structure serialization.
- Quantified **hedge fund positions** and **foreign central banks' holdings** of Government of Canada securities into dedicated time series to analyze the growing foreign investment in Canadian debt and their inherent risks.

EDUCATION

University of Waterloo — Bachelor of Mathematics

Jun 2024

Majors in Honours Statistics and Honours Computational Mathematics

Minor in Computing

- **Relevant courses:** Probability, Statistics, Stochastic Processes, Linear Algebra, Neural Networks, Microeconomics, Macroeconomics, Advanced Regression, Estimation & Hypothesis Testing, Databases, Algorithms, Optimization, Generalized Linear Models

PROJECTS

Canadian Bond Market Flow Report and Positions Tracker

Bank of Canada Internal Tool 

- Developed a **participant-level positions framework** by integrating primary and secondary markets data using Databricks and Python, enabling daily tracking of **outstanding bonds** and their **market flows** and **position changes**.
- Built an interactive auction-participation dashboard in PowerBI with configurable reporting windows to **monitor and regulate dealer and customer activity** and also gain insight for investor relations.
- Implemented entity-level aggregation and **trade-booking logic** in SQL to reconcile identity, positions, and flows across multiple data sources.

Spatial Models for Canadian Temperature Inference

[Github repo](#) 

- Compared spline regression, polynomial regression, and **generalized additive models** (GAMs) for **statistical spatial inference** of annual Canadian temperatures using R and the gamair package.
- Evaluated performance using generalized cross-validation and R^2 where GAM scored strongest with **17.83** and **0.89** respectively.

Tumor Modeling with HTC's and iNKT Cells

[Github repo](#) 

- Extended a **nonlinear differential-equation model** describing tumor and immune cell interactions by implementing numerical simulations in MATLAB.
- Analyzed equilibrium behavior and changes in system dynamics using **phase-plane**, **stability**, and **Hopf Bifurcation** analyses.

CERTIFICATIONS

Bloomberg Market Concepts

Bloomberg

- Portfolio Management, Fixed Income, FX, Equities, Options, Commodities, Terminal Basics

Microsoft Certified: Azure AI Fundamentals

Microsoft

- Recognized for fundamental knowledge of machine learning (ML) and artificial intelligence (AI) concepts and familiarity with technologies such as Kubernetes and other related basic cloud services.
- Applied concepts towards natural language processing (NLP), facial recognition, object detection, and image classification.